CSE 3330: Database Systems and File Structures

Fall 2026

As the instructor for this course, I reserve the right to adjust this syllabus and schedule in any way that serves the educational needs of the students enrolled in this course.
- Professor Ashraf Aboulnaga

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Instructor Information

Name

Professor Ashraf Aboulnaga

Office Location

ERB 622B

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Faculty Profile

University Profile: https://www.uta.edu/academics/faculty/profile?username=aboulnagaa
Personal Homepage: https://ashraf.aboulnaga.me

Office Hours

Tue 1:00 – 2:00 pm in ERB 622B

Teaching Assistant

Mr. Abdalla Mohamed

Email Address

axm0993@mavs.uta.edu

Office Hours

Tue and Thu 4:30 – 5:30 pm in ERB 514

Communication Guidelines

The preferred communication method for this course is email. Please include CSE 3330 in the subject line. The instructor or TA will respond within one business day.

Course Information

Section Information

CSE 3330 Section 001

Course Description

Course Title: Database Systems and File Structures

This is a first course in databases. The course covers the relational data model and its query languages, database design, transactions, database security, and performance. The course will also introduce database system internals and file/index data structures.

Prerequisites: C or better in each of the following: CSE 1325 and CSE 3318

Time and Place of Class Meetings

Tue and Thu 11:00am to 12:20pm in Room NH 111

Time Zone

This course operates on Central Time. All times listed for class meeting times, exams, and assignment deadlines are in Central Time (CT).

Classroom/Lecture Recording Policy

Faculty maintain the academic right to determine whether students are permitted to record classroom and online lectures. Recordings of classroom lectures, if permitted by the instructor or pursuant to an ADA accommodation, may only be used for academic purposes related to the specific course. They may not be used for commercial purposes or shared with non-course participants except in connection with a legal proceeding.

Recording of classroom and online lectures in this course is allowed.

Student Learning Outcomes

By the end of this course, you will be able to:

1. **Querying:** Given a relational database schema and the specification of a query in English, students will be able to compose correct queries in relational algebra or SQL (Structured Query Language) to retrieve the data the meets the query specification.
2. **Database Design:** Given the description of a database in English, students will be able to draw an ERD (Entity-Relationship Diagram) that correctly models the database and translate the ERD to CREATE TABLE statements in SQL.
3. **Functional Dependencies and Normalization:** Students will be able to describe the role of functional dependencies in maintaining database integrity and correctly use functional dependencies to change a database schema to Boyce-Codd Normal Form.
4. **Programming:** Students will be able to write programs that connect to a database system and effectively modify or retrieve data in a database managed by this system.
5. **Transactions:** Students will be able to accurately describe the guarantees offered by database systems to user transactions and identify the effect of these guarantees on their programs.

Course Materials and Technology

Textbook Information

Recommended Textbook

The following textbook is optional but recommended:

Ramez Elmasri and Shamkant Navathe, “Fundamentals Of Database System,” 7th edition, Pearson, 2016. ISBN: 9780133970777. Price in UTA Bookstore: $253.50 (new)

Technology & Equipment Requirements

- The primary mode of communication will be Canvas
- We will take attendance using aPlus+ Attendance in Canvas
- We may occasionally use Teams
- If there is a need for on-line exams, we will use the Respondus lockdown browser
- The homeworks and programming assignment require access to the MySQL database management system, which you must install on your computer. Instructions for installation will be provided.

Visit the OIT Services page for a list of Applications and Software available through UTA.

Visit the UTA Libraries Technology page for a list of items that can be checked out or used at the library.

Assignments & Exams

Exams

- Exam 1 – September 10
  Exam 2 – October 8
  Exam 3 – November 5
  Exam 4 – December 3 (Final exam week)
- The first three exams will be held in class and will take up the full class time.
- The last exam will be held during final exam week. You will have more time, but the exam will be similar in length to the first three exams.
- Pencil/pen and paper, no calculators, computers, or other aids.
- The exams are worth 75% of your final grade. The lowest-scoring exam for each student will be dropped. The remaining exams will be worth 25% each.
- The exams will cover the material since the beginning of the course, with a strong emphasis on the material covered since the most recent exam.

Homeworks

- Homework 1 – Released August 20, Due September 4
  Homework 2 – Released September 15, Due October 2
  Homework 3 – Released October 13, Due October 30
  Homework 4 – Released November 10, Due November 20
- All homeworks are due on a Friday at 11:59pm. No extensions will be given.

- The homeworks will involve pencil/pen and paper problems and also computer-based problems done on the MySQL database system. Homework 4 will involve Python programming and connecting to the MySQL database system.
- Each homework is worth 5% of your final grade.

Attendance

- We will take attendance using aPlus+ Attendance in Canvas
- Attendance is worth 5% of your final grade

Grading Information

Assignments	Values (%)
Exam 1*	25%
Exam 2*	25%
Exam 3*	25%
Exam 4*	
Homework 1	5%
Homework 2	5%
Homework 3	5%
Homework 4	5%
Attendance	5%
	Total: 100%

* Lowest scoring exam will be dropped and the remaining three will be worth 25% each.

Students are expected to track their performance throughout the semester, which Canvas facilitates, and seek guidance from available sources, including the instructor, if their performance drops below satisfactory levels. Refer to the Student Support Services section below.

Final Grade Calculations

Earned % Range	Letter Grade
90-100%	A
80-89%	B
70-79%	C
60-69%	D
0-59%	F

Late Work Policy

Late homework will not be accepted.

Make-Up Exams Policy

There will be no make-up exams. If you miss an exam, that exam will be the one whose score is dropped. There are no makeups for pop quizzes.

Extra Credit Policy

In general, this course does not have extra credit. However, if there is a reason and opportunity to create extra credit assignments, the instructor will create them and they will count towards the 5% Attendance portion of the grade

Grade Grievance Policy

Any appeal of a grade in this course must follow the procedures and deadlines for grade-related grievances as published in the current University Catalog: Grades and Grading Policies.

Course and University Policies

Attendance Policy

Students should review the University Class Attendance Policies on the Class Attendance Policies page. The following attendance policy will be applied in this course.

I will take attendance in lectures using aPlus+ Attendance in Canvas. Attendance is worth 5% of your final grade.

Generative AI Use in This Course

The use of Generative AI (GenAI) in course assignments and assessments must align with the guidelines established by the instructor. Unauthorized use of GenAI could result in breaches of academic integrity. Instructors are responsible for clearly delineating the permissible uses of GenAI in their courses, underscoring the importance of responsible and ethical application of these tools.

Community Standards within the Office of the Dean of Students articulate the university's stance on academic integrity and scholastic dishonesty. These standards extend to the use of GenAI. Unauthorized or unapproved use of GenAI in academic work falls within the scope of these policies and will be subject to the same disciplinary procedures.

As the instructor for this course, I have adopted the following policy on student use of GenAI.

Students may use GenAI tools freely to assist in understanding course content and creating homework solutions. It is expected that students will engage with these tools ethically and responsibly, ensuring that their use of GenAI contributes to a deeper understanding of the subject matter and the development of relevant competencies. Students are reminded that GenAI is not available in exams, which constitute the bulk of the grade in this course.

Institutional Policies

UTA students should review the University Catalog and the Syllabus Institutional Policies page for institutional policies and contact the specific office with any questions. The institutional information includes the following policies, among others:

- Drop Policy
- Disability Accommodations
- Academic Integrity

- Electronic Communication

UTA Honor Code

UTA students are expected to adhere to and observe standards of conduct compatible with the University's functions as an educational institution and live by the University of Texas at Arlington's Honor Code. It is the policy of The University of Texas at Arlington to uphold and support standards of personal honesty and integrity for all students consistent with the goals of a community of scholars and students seeking knowledge and responsibility.

Student Support Services

Student Services Page

The Student Services page provides links to many resources available to UTA students, including:

- Academic Success
- Counseling and Psychological Services (CAPS)
- Health Services
- Students with Disabilities
- Veteran Services

Students are also encouraged to check out Career Center resources to enhance their career-readiness, find student employment, search for internships, and more. We encourage Major Exploration and the use of Experiential Major Maps to keep students on track for graduation. Refer to the Graduation Help Desk for more details.

Accessibility of Course Materials

Some course materials, such as PDFs of musical scores, technical drawings, graphs, blueprints, design plans, or artworks (common in fields like drawing, painting, or construction drafting), may not fully comply with all Web Content Accessibility Guidelines (WCAG) requirements.

The University of Texas at Arlington is dedicated to ensuring all students have equal access to information. If you experience any accessibility barriers with course materials, please know that accommodations are available. You can get assistance through the Student Access and Resource (SAR) Center or by contacting your instructor directly. Please don't hesitate to reach out if you need help.

Online Academic Success Guide

Visit the Online Academic Success Guide to explore a list of helpful tips and resources to help you succeed in your online journey.

UTA Health and Wellbeing Resources

UT Arlington is committed to the safety, success, and well-being of our students. To support our community, UTA has established a Community Advocacy, Response, and Engagement (CARE) Team, a dedicated group of campus professionals responsible for helping students who could

benefit from academic, emotional, or psychological support, as well as those presenting risks to the health or safety of the community. If you know of someone experiencing challenges, appearing distressed, needing resources, or causing a significant disruption to the UTA community, please submit a CARE Referral by visiting the CARE Team page. You may also submit a referral for yourself if you would like additional support.

UTA students also have access to virtual, on-demand emotional support, appointment-based counseling, advanced psychiatric care, and more. For more information, visit TimelyCare.

NOTE: If a person's behavior poses an immediate threat to you or someone else, contact UTA Police at 817-272-3003 or dial 911. If you or someone you know needs to speak with a crisis counselor, please reach out to the MAVS TALK 24-hour Crisis Line at 817-272-8255 or the National Suicide and Crisis Lifeline at 988.

Librarian to Contact

Each academic unit has access to Librarians by Academic Subject who can assist students with research projects, tutorials on plagiarism, citation references, as well as support with databases and course reserves.

Course Schedule

Module and Dates	Topics	Textbook Chapters	Additional Chapters	Assignments Due
8/18	Introduction	1, 2		
8/20	Relational Model	5		HW 1 Released
8/25	Relational Algebra	8		
8/27	Relational Algebra	8		
9/1	Relational Algebra	8		
9/3	SQL	6, 7		
Fri 9/4	--			HW 1 Due
9/8	SQL	6, 7		
9/10	**Exam 1**			
9/15	SQL	6, 7		HW 2 Released
9/17	SQL	6, 7		
9/22	SQL	6, 7		
9/24	SQL	6, 7		
9/29	Entity-Relationship Modeling	3, 4, 9		
10/1	Entity-Relationship Modeling	3, 4, 9		
Fri 10/2	--			HW 2 Due
10/6	Entity-Relationship Modeling	3, 4, 9		
10/8	**Exam 2**			
10/13	Entity-Relationship Modeling	3, 4, 9		HW 3 Released
10/15	Functional Dependencies and Normalization	14		
10/20	Functional Dependencies and Normalization	14		
10/22	Functional Dependencies and Normalization	14		
10/27	Database Programming		10, 11	
10/29	Transactions	20	21, 22	
Fri 10/30	--			HW 3 Due
11/3	Transactions	20	21, 22	
11/5	**Exam 3**			
11/10	Transactions	20	21, 22	HW 4 Released
11/12	Database Security	30		
11/17	Data Storage and Indexing	16, 17		
11/19	Query Processing and Optimization	18	19	
Fri 11/20	--			HW 4 Due
11/24	Performance Tuning		16, 17	
12/1	Beyond Relational Databases		24, 25, 26, 29	
12/3	**Exam 4 (Final exam week)**			